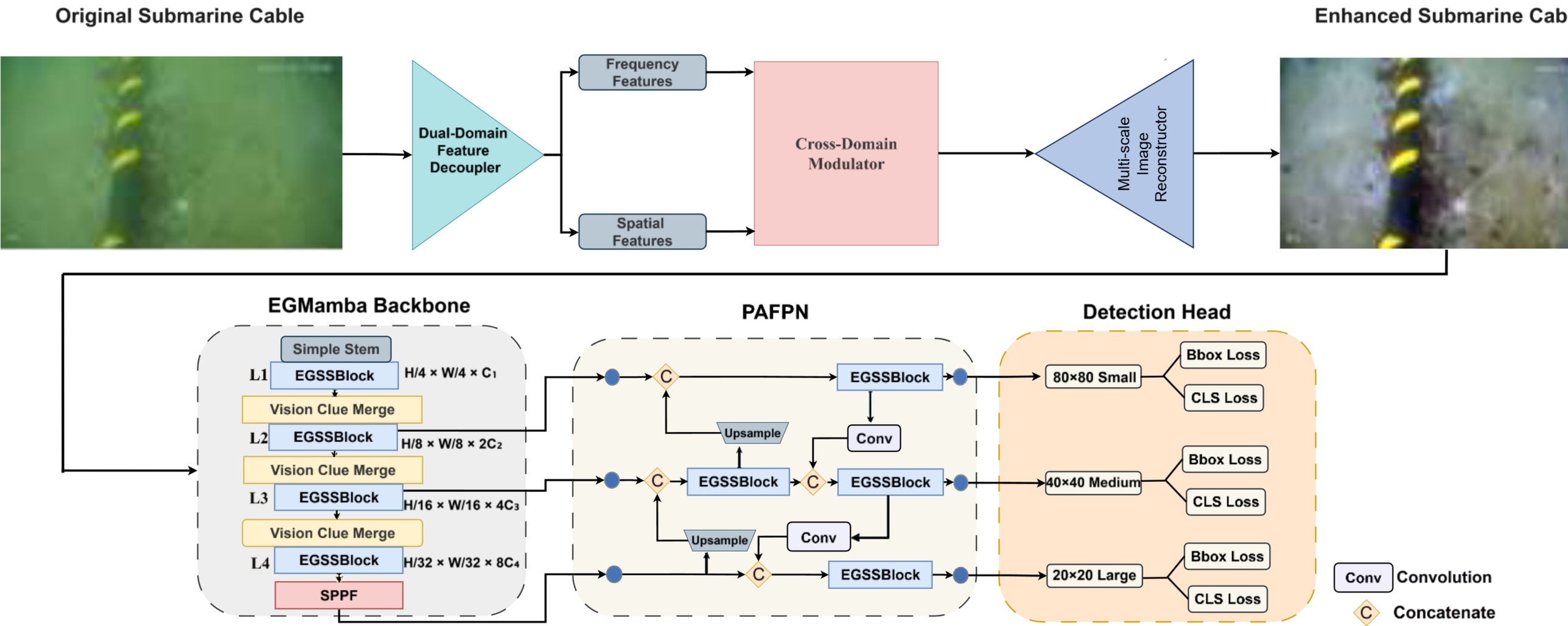
Original Submarine Cable
Enhanced Submarine Cab
Dual-Domain Feature Decoupler
Frequency Features
Spatial Features
Cross-Domain Modulator
Multi-scale Image Reconstructor
EGMamba Backbone
Simple Stem
L1
EGSSBlock
H/4 × W/4 × C1
Vision Clue Merge
L2
EGSSBlock
H/8 × W/8 × 2C2
Vision Clue Merge
L3
EGSSBlock
H/16 × W/16 × 4C3
Vision Clue Merge
L4
EGSSBlock
H/32 × W/32 × 8C4
SPPF
PAFPN
C
EGSSBlock
Upsample
Conv
C
EGSSBlock
C
EGSSBlock
Upsample
Conv
C
EGSSBlock
Detection Head
80×80 Small
Bbox Loss
CLS Loss
40×40 Medium
Bbox Loss
CLS Loss
20×20 Large
Bbox Loss
CLS Loss
Conv Convolution
C Concatenate